

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

School of Computing — Department of Computer Science & Engineering

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Course: 21CSC203P — Advanced Programming Practice	Department: Computer Science and Engineering

Unit II | Session 3 | SLO 2: Polymorphism (Compile-time & Runtime) (Code: 2032)

Q: Activity: Write a program demonstrating both Compile-Time Polymorphism (Method Overloading) and Runtime Polymorphism (Method Overriding via Dynamic Method Dispatch) using geometric shapes.

Ans: Java Implementation:

```
class ShapeCalculator {
    // Method Overloading (Compile-time polymorphism)
    double area(double radius) {
        return Math.PI * radius * radius;
    }
    double area(double length, double width) {
        return length * width;
    }
}

class Shape {
    void draw() {
        System.out.println("Drawing generic shape");
    }
}

class Circle extends Shape {
    @Override
    void draw() {
        System.out.println("Drawing Circle");
    }
}

public class PolymorphismDemo {
    public static void main(String[] args) {
        ShapeCalculator calc = new ShapeCalculator();
        System.out.println("Circle Area: " + calc.area(4.0));
        System.out.println("Rectangle Area: " + calc.area(4.0, 6.0));

        // Dynamic Method Dispatch (Runtime Polymorphism)
        Shape myShape = new Circle();
        myShape.draw();
    }
}
```